

EDUCATION

The University of Auckland

Auckland, New Zealand

DOCTOR OF PHILOSOPHY (PHYSICS)

September 2021 - August 2025

- Supervisors: Prof. Jan (JJ) Eldridge & Assoc. Prof. Nicholas Rattenbury
- Thesis: *The Next Generation: Refining the Standard Population Synthesis Model to Better Predict Stellar Population Properties*

The University of Auckland

Auckland, New Zealand

MASTER OF SCIENCE (PHYSICS)

March 2020 - March 2021

- Supervisor: Prof. Jan (JJ) Eldridge
- Thesis: *The Missing Mergers: Modelling the observed binary neutron-star population and their mergers*
- Awarded with **Honours (First Class)**

The University of Auckland

Auckland, New Zealand

POSTGRADUATE DIPLOMA IN SCIENCE (PHYSICS)

March 2019 - February 2020

- Supervisor: Assoc. Prof. Nicholas Rattenbury
- Dissertation: *The University of Auckland Ground Station One*
- Technical report: *The University of Auckland Ground Station One: System Overview*
- Technical report: *Ground Segment Support for APSS-I* (contains non-public information, and cannot be distributed)

The University of Auckland

Auckland, New Zealand

BACHELOR OF SCIENCE

February 2016 - November 2018

- Majored in Physics and Computer Science

PROFESSIONAL EXPERIENCE

2025- **Technician**, The University of Auckland

AWARDS, FELLOWSHIPS, & GRANTS

- 2021 **University of Auckland Doctoral Scholarship**, The University of Auckland
- 2019 **First in Course Award (PHYSICS 746: Relativistic Quantum Mechanics and Quantum Field Theory)**, The University of Auckland

PRESENTATIONS

* *presenting author*; + *mentored undergraduate*

CONTRIBUTED PRESENTATIONS

Richards, S*, Eldridge, J, Ghodla, S, Briel, M. 2026. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material*. Oral presentation: National Astronomy Meeting 2026, Birmingham, United Kingdom.

Richards, S 2024. *A fate written in stardust: Probing the interiors of interacting binary stars*. Oral presentation: The 2nd Faculty of Science Pacific Research Symposium, Auckland, New Zealand.

Richards, S, Burn, L. 2024. *Beyond the Bias: Student insights on equity from Auckland to Exeter*. Colloquium presentation: The University of Auckland Department of Physics, Auckland, New Zealand.

Richards, S*, Eldridge, J, Briel, M, Ghodla, S. 2024. *The variable evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material*. Oral presentation: Annual Scientific Meeting of the Astronomical Society of Australia (online).

Richards, S*. 2022. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations*. Oral presentation: Supernovae in the Gravitational Wave Detection Era, Melbourne, Victoria, Australia.

TEACHING EXPERIENCE

2026	PHYSICS 753 , Course Coordinator	<i>The University of Auckland</i>
2023-2025	PHYSICS 102 , Tutor	<i>The University of Auckland</i>
2021-2023	Tuākana Programme , Tutor	<i>The University of Auckland</i>
2018-2025	ASTRO 100 / PHYSICS 107 , Graduate Teaching Assistant	<i>The University of Auckland</i>

RESEARCH EXPERIENCE

University of Auckland - Department of Physics

SUPERVISOR: ASSOC. PROF. NICHOLAS RATTENBURY

- Research Assistant: University of Auckland Ground Station One

Auckland, NZ

June 2023 - December 2023

University of Auckland - Department of Physics

SUPERVISOR: PROF. JAN (JJ) ELDRIDGE

- Research Assistant

Auckland, NZ

April 2021 - September 2021

University of Auckland - Department of Physics

SUPERVISOR: ASSOC. PROF. NICHOLAS RATTENBURY

- Summer Research Scholar: "Ground Segment Support for APSS-I"
- This report is non-distributable due to a non-disclosure agreement

Auckland, NZ

December 2019 - February 2020

OUTREACH & PROFESSIONAL DEVELOPMENT

ACADEMIC CITIZENSHIP AND SERVICE

2025	Department of Physics Equity Committee , Chair	<i>The University of Auckland</i>
2023-	Department of Physics Equity Committee , Member	<i>The University of Auckland</i>

SOCIETIES

2021	Post-Graduate Students Association , Welfare Vice President	<i>The University of Auckland</i>
2021	Physics Association of the University of Auckland , Vice President	<i>The University of Auckland</i>
2020	Physics Association of the University of Auckland , Treasurer	<i>The University of Auckland</i>
2018	Physics Association of the University of Auckland , Committee Member	<i>The University of Auckland</i>

PROFESSIONAL DEVELOPMENT

First Aid Certificate, September 2025 (expiry September 2027)

PUBLICATIONS

PUBLISHED

Richards, S., Eldridge, J., Briel, M., Stevance, H., Willcox, R. 2023. *New constraints on the Bray conservation-of-momentum natal kick model from multiple distinct observations*. Monthly Notices of the Royal Astronomical Society.

Stevance, H., Ghodla, S., **Richards, S.**, Eldridge, J., Briel, M., Tang, P. 2023. *VFTS 243 as predicted by BPASS models*. Monthly Notices of the Royal Astronomical Society.

Richards S., Eldridge J., Briel M., Ghodla S. 2025. *The evolution of accretor stars in binary systems due to accretion of increasingly helium-rich material*. Monthly Notices of the Royal Astronomical Society.

IN REVIEW

IN PREP

Mandhai S., Breton R., Kay S., **Richards, S.** 2026. *The orbital evolution of NSNS in Milky Way like galaxies.*